Player Tracking in Sports: Multi-View Tracking and 3D Reconstruction

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1 Introduction

Vision-based tracking provides a scalable alternative to expensive sports analytics hardware, but basketball's rapid movements, mutual occlusions, motion blur, and specular floor reflections pose severe challenges.

To address these challenges, modern pipelines have largely abandoned classical background subtraction [3] in favor of tracking-by-detection paradigms. Real-time localization is typically driven by single-stage detectors like the YOLO family [5]. To link these spatial detections temporally, SORT [1] established a baseline using Kalman filtering, which was later improved to handle severe occlusions using deep appearance embeddings (DeepSORT [6, 8]) and low-confidence secondary associations (ByteTrack [7]). Finally, multi-camera configurations [3] resolve remaining overlaps by applying epipolar geometry and triangulation to lift 2D sequences into 3D space.

Building on these foundations, we present an end-to-end pipeline for multi-view player tracking and 3D reconstruction. We systematically evaluate these strategies, demonstrating that a domain-adapted YOLO26 with class-agnostic NMS [4] combined with appearance-based DeepSORT heavily outperforms classical baselines in reducing identity switches. Finally, we reconstruct 3D trajectories across three calibrated cameras using SVD-based triangulation, utilizing Kalman smoothing to suppress detection noise.

2 Detection

This section describes the three detection approaches evaluated in this work, progressing from a classical background subtraction baseline to a fine-tuned deep learning model with multi-pass inference.

2.1 Background Subtraction (MOG2)

As a baseline, we employ Mixture of Gaussians (MOG2) background subtraction [3]. To mitigate illumination variations, frames are preprocessed using Contrast Limited Adaptive Histogram Equalization (CLAHE) [9]. The generated foreground mask is subsequently refined via morphological operations and connected component analysis.

Despite these enhancements, this baseline suffers from four primary failure modes. First, floor specularities create intense reflections and shadows that evade MOG2's color-based shadow detection. Then, temporarily stationary players are erroneously integrated into the background model, and dynamic backgrounds, such as moving spectators, introduce persistent false positives.

2.2 Pre-Trained YOLOv11

To overcome MOG2 limitations, we deploy a YOLOv11 architecture [5] pre-trained on the COCO dataset. Transitioning to this object detection paradigm substantially improves accuracy and mitigates the limitations of the background subtraction baseline.

However, two distinct domain gaps persist: the sports ball is entirely absent from the pre-trained taxonomy, and players and referees are uniformly classified as "person", precluding downstream identity distinction.

2.3 Fine-Tuned YOLOv26 with Multi-Pass Inference

To bridge these domain gaps, we adopt the YOLOv26 architecture and fine-tune its backbone on a domain-specific annotated dataset. The model is trained to recognise each player individually, as well as the ball, so identity is resolved directly by the detector.

However, because the ball occupies a minimal pixel area, standard single-pass inference downsamples it too aggressively for reliable localization. We circumvent this spatial resolution trade-off via a dual-pass inference strategy: a primary pass optimized for player detection, and a parallel high-resolution pass configured with a permissive confidence threshold specifically tailored to isolate the ball. The resulting detections are subsequently unified into a single composite output.

Class-Agnostic NMS A player may be detected multiple times under different identities, which ordinary non-maximum suppression cannot merge since it only compares same-identity boxes. We instead apply a class-agnostic variant [4], suppressing by spatial overlap alone to keep one box per player, yielding the final detections passed to the tracker.

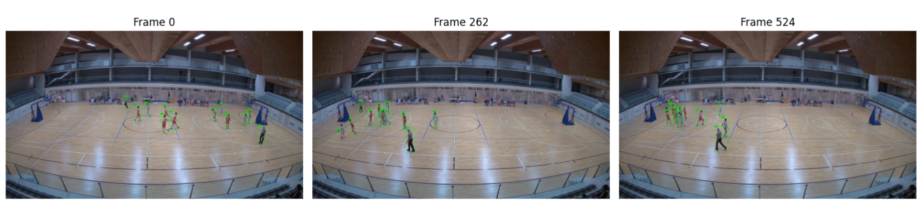

Fig. 1. Sample detection output from fine-tuned YOLOv26 with class-agnostic NMS on cam_13 (three selected frames).

3 Tracking

This section progresses from the SORT baseline to a DeepSORT tracker with several robustness mechanisms, and finally to a label resolution step.

3.1 Simple Online and Realtime Tracking (SORT)

As a baseline tracker, we adopt Simple Online and Realtime Tracking (SORT) [1]. Each track is propagated forward by a Kalman filter under a constant-velocity motion model, and the predicted boxes are associated to the new detections via the Hungarian algorithm, using spatial overlap (IoU) as the matching cost.

Its main limitation is that it relies on proximity alone, with no notion of appearance: when two players cross, their predicted boxes overlap and the tracker frequently swaps their identities.

3.2 DeepSORT

To minimize identity swaps, our final tracking framework utilizes DeepSORT [6], which augments the traditional Kalman filter motion model with appearance cues. For each detection, a lightweight OSNet architecture [8] extracts an appearance embedding, appended to a rolling gallery for each active trajectory. Gating and association proceed via a matching cascade that prioritizes appearance similarity over spatial overlap.

On top of this base we introduce four mechanisms that make tracking more robust to the occlusions, missed detections, and noise typical of our footage.

Confidence-Aware Kalman Update (NSA) Standard Kalman filtering weighs all bounding box updates uniformly during state correction. To minimize tracking jitter, we scale the measurement covariance by the detection confidence, ensuring high-confidence detections exhibit greater influence than uncertain ones.

Low-Confidence Recovery (ByteTrack) Low-confidence detections often represent genuine players experiencing transient occlusion or motion blur. Adopting the ByteTrack [7] paradigm, unmatched tracks are given a second-pass association against these low-confidence boxes based strictly on spatial overlap. This recovery mechanism maintains track continuity through partial occlusions; however, to prevent gallery corruption from noisy bounding boxes, these detections are excluded from appearance embedding updates.

Track Resurrection When a trajectory is lost due to prolonged occlusion, standard trackers terminate the track, causing a re-emerging player to receive a new identity. We retain appearance galleries of recently terminated tracks for a short temporal window and restore an original identity when a new detection closely matches a retained gallery.

Coasting When a track is momentarily left unmatched, it continues to report its Kalman-predicted box so that short gaps are bridged, provided this prediction remains within the frame and does not overlap a box already assigned to another track.

3.3 Label Resolution

Because the fine-tuned detector assigns unique class labels to individual player identities, a single trajectory may accumulate conflicting class predictions across frames. To resolve this, each track is assigned its confidence-weighted majority label, with any identity claimed by two tracks kept by the more confident one and the other taking its next-best label. The ball class is exempt from this arbitration procedure, as it represents a global category rather than a mutually exclusive unique identity.

Fig. 2. Sample tracking output from DeepSORT with resolved identities on cam_13 (three selected frames).

4 3D Reconstruction

Having recovered the per-camera 2D trajectories, we lift them into a single 3D representation of the scene shared by all three cameras.

4.1 Camera Calibration

Intrinsic camera matrices are known a priori. Extrinsic parameters are subsequently estimated by solving the Perspective-n-Point (PnP) problem, mapping manually annotated 2D image coordinates of court landmarks from the initial frame to their corresponding 3D world coordinates. Lens distortion coefficients are integrated directly within the PnP optimization framework to ensure simultaneous rectification and alignment.

4.2 Triangulation

Detections are grouped by identity and frame index across the three cameras, then undistorted using the estimated intrinsics. Players and the ball are reconstructed separately, since they move differently.

Players Players are assumed to stand on the court floor. Each camera's foot pixel is back-projected into a ray, and its intersection with the floor gives a ground position. This constraint lets us localise a player even from a single view, which helps during occlusions. Estimates from several cameras are merged by discarding outliers and averaging the rest.

Ball The ball is airborne, so it cannot be tied to the floor. Instead, its position is triangulated from the back-projected rays of all cameras that see it, requiring at least two views. The rays are intersected with the Direct Linear Transform (DLT), which stacks the per-camera constraints into a homogeneous linear system and solves it via Singular Value Decomposition (SVD). Views with a high reprojection error are discarded, and the point is re-triangulated from the rest.

4.3 Trajectory Smoothing

The triangulated trajectories still contain noise from detection jitter, calibration error, and outliers. We smooth each one with a Kalman filter run forward in time, followed by a Rauch–Tung–Striebel (RTS) backward pass, so every point is corrected using both past and future observations. Points that jump implausibly are rejected as outliers, and a trajectory is split whenever a gap grows too long, to avoid extrapolating across long occlusions.

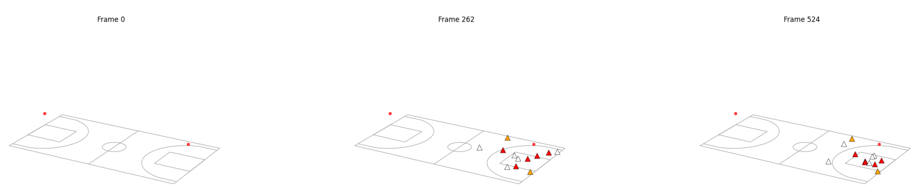

Fig. 3. Reconstructed 3D player and ball positions projected onto the court plane after RTS smoothing.

5 Evaluation Metrics

We evaluate each stage of the pipeline separately: detection quality, tracking quality, and the accuracy of the final 3D reconstruction.

5.1 Detection Metrics

Detection performance is evaluated using standard precision, recall, and the F_1-score, with true positives defined at an Intersection over Union (IoU) threshold of 0.5.

5.2 Tracking Metrics

We evaluate tracking performance using IDF1 and HOTA [2]. The IDF1 score measures identity consistency over time by computing the harmonic mean of identity precision and identity recall.

Our primary metric, HOTA, balances detection and association accuracy by factoring in both detection accuracy (DetA) and association accuracy (AssA). The final HOTA score is computed as the geometric mean of DetA and AssA, averaged over a range of localization thresholds.

We favor HOTA over traditional MOT metrics like MOTA, which are heavily biased toward detection errors and fail to adequately reflect tracking continuity.

5.3 3D Geometry Metrics

We assess 3D geometric fidelity using reprojection and trajectory errors. Reprojection error measures the pixel distance between a triangulated 3D point and its original 2D detection when projected back onto each observing camera's image plane, serving as an indicator of geometric consistency; we report its mean, median, root-mean-square error (RMSE).

Trajectory error evaluates tracking accuracy by reprojecting the 3D tracks back into the camera views and comparing them per identity against the ground-truth 2D trajectories across all shared frames. This is quantified via the Average Displacement Error (ADE) to measure mean spatial drift, the Final Displacement Error (FDE) to capture long-term accumulation errors, the Median Trajectory Error (MTE) to provide robustness against outlier frames, and the total number of trajectory fragments to index coverage gaps.

6 Results

We summarise here the final tracking and trajectory results. The full set of detection, tracking, and geometry tables, together with the per-stage qualitative outputs and all pipeline parameters, is reported in the appendix (Appendices B–D).

6.1 Tracking

Table 1 reports the metrics of our final tracker, DeepSORT, across the three cameras.

Table 1. Final tracking results (DeepSORT).

Camera	IDF1	HOTA	DetA	AssA
cam_13	0.569	0.434	0.469	0.405
cam_2	0.610	0.480	0.453	0.511
cam_4	0.638	0.521	0.553	0.495

DeepSORT maintains identities reasonably well across all views, with the best scores on cam_4. The similar DetA and AssA values indicate that detection recall, rather than association, is the main factor limiting tracking quality on this short clip.

6.2 Trajectory

Table 2 reports the error of the reconstructed 3D tracks projected back onto the annotated 2D trajectories.

Table 2. Trajectory error of projected 3D tracks vs. ground-truth 2D annotations.

Camera	ADE (px)	FDE (px)	MTE (px)	Fragments
cam_13	976.8	849.8	1047.5	2
cam_2	1066.0	1279.1	1181.4	5
cam_4	1838.2	1689.1	1899.9	5

The trajectory errors are large across all views, with the low fragment count showing that the tracks still cover the annotated span almost without gaps. The dominant source of error is calibration accuracy: a bias in the estimated camera pose propagates directly into a constant positional offset in the reconstructed trajectories.

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A Dataset

The pipeline is evaluated on a short basketball sequence recorded at the Sanbapolis facility by three synchronised cameras (cam_13, cam_2, and cam_4). The clip spans 525 frames at 25 fps (about 21 seconds) and covers a full FIBA court of 28×15 m.

Each camera's intrinsic parameters are provided. The extrinsics are estimated by solving the Perspective-n-Point problem from manually selected court landmarks on the first frame, with lens distortion modelled directly inside the solver.

For evaluation, ground-truth 2D bounding boxes are annotated on every fifth frame. The detector recognises the ball as a single class and each player as its own identity class; it was fine-tuned on a separate dataset from the same domain, covering different actions of the same game.

B Pipeline Parameters

The following tables list the parameters used at each stage of the pipeline.

Table 3. MOG2 background-subtraction parameters.

Parameter	Value
Learning rate	-1 (adaptive)
History (frames)	1000
Variance threshold	25
Opening kernel (px)	7
Closing kernel (px)	11
Min. contour area (px^2)	1500
Max. contour area (px^2)	200000

Table 4. YOLO detection parameters (pre-trained, fine-tuned multi-pass, and NMS).

Parameter	Value
Pre-trained YOLOv11	
Confidence threshold	0.4
Built-in NMS IoU	0.45
Inference size (px)	640
Fine-tuned YOLOv26 (multi-pass)	
Player confidence	0.3
Player low-conf band	0.15
Ball confidence	0.2
Player inference size (px)	1280
Ball inference size (px)	1600
Class-agnostic NMS	
IoU suppression threshold	0.75

Table 5. Tracking parameters (SORT and DeepSORT).

Parameter	Value
SORT	
Max. IoU distance	0.8
Max. age (frames)	60
Confirmation hits	2
DeepSORT	
Max. IoU distance	0.8
Max. appearance distance	0.2
Max. age (frames)	60
Confirmation hits	2
Feature budget per track	100

Table 6. 2D trajectory smoothing, triangulation, and 3D smoothing parameters.

Parameter	Value
2D trajectory smoothing	
Max. interpolated gap (frames)	5
EMA weight α	0.5
Triangulation	
Ground inlier gate (mm)	600
Ball reprojection gate (px)	30
Minimum views	2
3D trajectory smoothing	
Position noise std (mm)	30
Velocity noise std (mm/frame)	60
Measurement noise std (mm)	80

C Detailed Results

Table 7. Detection results across cameras (IoU threshold = 0.5).

Method	Camera	Precision	Recall	F_1	Mean IoU
MOG2	cam_13	0.471	0.262	0.337	0.651
	cam_2	0.345	0.408	0.374	0.663
	cam_4	0.328	0.293	0.309	0.700
YOLO pre-trained	cam_13	0.834	0.602	0.699	0.748
	cam_2	0.819	0.781	0.800	0.785
	cam_4	0.927	0.807	0.862	0.808
YOLO fine-tuned	cam_13	0.872	0.732	0.796	0.758
	cam_2	0.835	0.761	0.796	0.757
	cam_4	0.923	0.822	0.870	0.796

Table 8. Identity tracking metrics (IoU threshold = 0.5).

Tracker	Camera	IDP	IDR	IDF1
SORT	cam_13	0.660	0.514	0.590
	cam_2	0.743	0.607	0.668
	cam_4	0.768	0.605	0.679
DeepSORT	cam_13	0.588	0.533	0.569
	cam_2	0.617	0.603	0.610
	cam_4	0.670	0.605	0.638

Table 9. HOTA tracking metrics.

Tracker	Camera	HOTA	DetA	AssA	LocA
SORT	cam_13	0.438	0.484	0.398	0.797
	cam_2	0.517	0.515	0.522	0.794
	cam_4	0.542	0.575	0.514	0.829
DeepSORT	cam_13	0.434	0.469	0.405	0.769
	cam_2	0.480	0.453	0.511	0.761
	cam_4	0.521	0.553	0.495	0.798

Table 10. Reprojection error of triangulated 3D points.

Camera	Mean (px)	Median (px)	RMSE (px)	Acc@5px	Acc@10px	Acc@20px
cam_13	51.3	28.9	87.7	0.135	0.221	0.349
cam_2	140.0	94.5	278.7	0.028	0.047	0.094
cam_4	135.0	50.0	296.2	0.168	0.214	0.295

Table 11. Trajectory error of projected 3D tracks vs. ground-truth 2D annotations.

Camera	ADE (px)	FDE (px)	MTE (px)	Fragments
cam_13	976.8	849.8	1047.5	2
cam_2	1066.0	1279.1	1181.4	5
cam_4	1838.2	1689.1	1899.9	5

D Qualitative Outputs

The figures below are the per-stage visual outputs produced by the pipeline notebook on cam_13.

Fig. 4. Raw input frame.

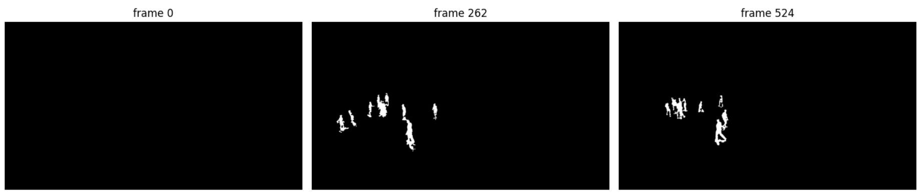

Fig. 5. MOG2 foreground mask.

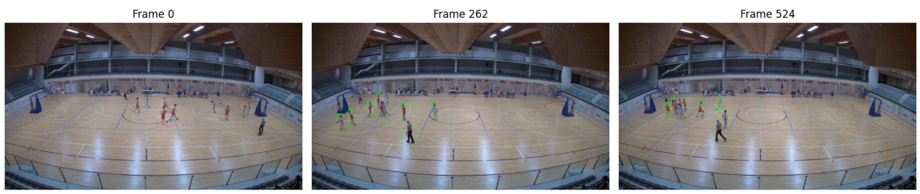

Fig. 6. MOG2 detections overlaid on the frame.

Fig. 7. Pre-trained YOLOv11 detections.

Fig. 8. Fine-tuned YOLO merged player and ball detections.

Fig. 9. Detections after class-agnostic NMS.

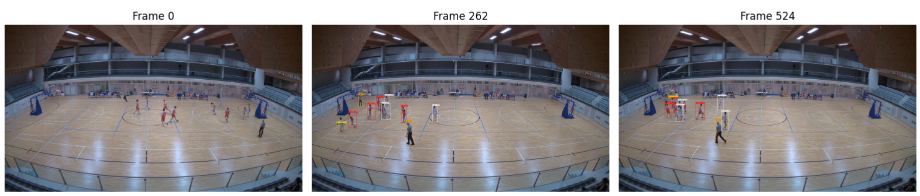

Fig. 10. SORT tracks.

Fig. 11. SORT tracks after label resolution.

Fig. 12. DeepSORT tracks.

Fig. 13. DeepSORT tracks after smoothing and label resolution.

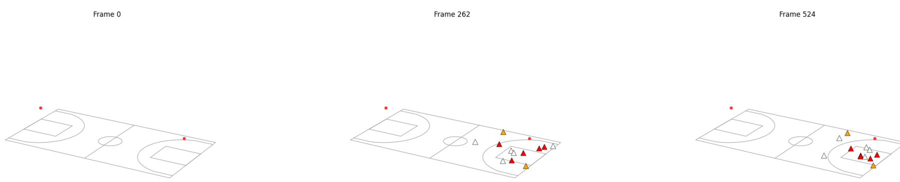

Fig. 14. Triangulated 3D reconstruction (top view).

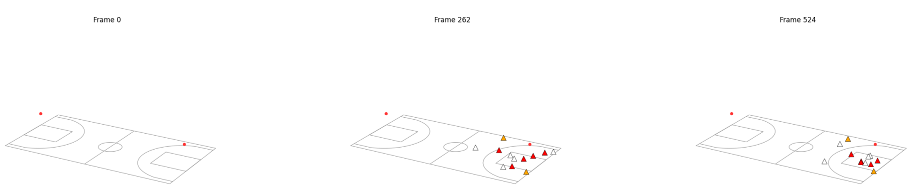

Fig. 15. Smoothed 3D reconstruction (top view).